

Oblique triangle cases solved using the Law of Cosines.

Case 3 (SAS) and Case 4 (SSS)

I. The Law of Cosines

The **Law of Cosines** says that in any triangle ABC , these three relationships exist among the angles A , B , and C and their opposite sides a , b , c :

Sides:

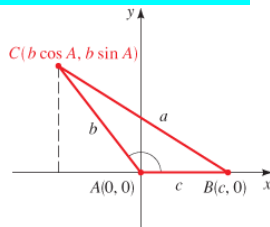
$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$b^2 = a^2 + c^2 - 2ac \cos B$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

Angles:

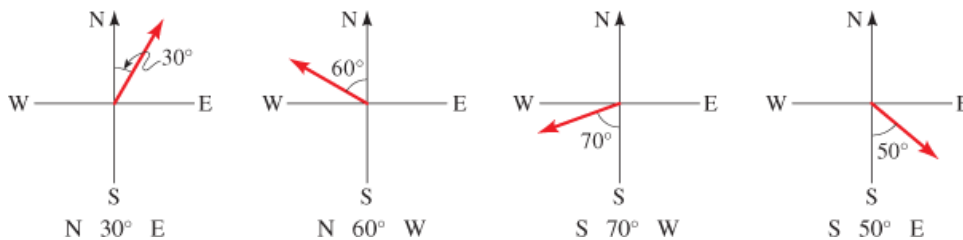
$$\cos A = \frac{b^2 + c^2 - a^2}{2bc}, \quad \cos B = \frac{a^2 + c^2 - b^2}{2ac}, \quad \cos C = \frac{a^2 + b^2 - c^2}{2ab}$$



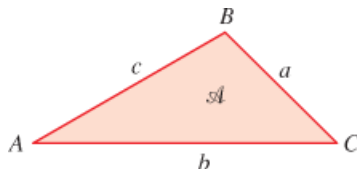
In general, the Law of Cosines says that the square of any side of a triangle is equal to the sum of the squares of the other two sides, minus twice the product of those two sides times the cosine of the included angle.

II. Navigation: Heading and Bearing

In navigation a direction is often given as a **bearing**, that is, as an acute angle measured from due north or due south.



III. The Area of a Triangle

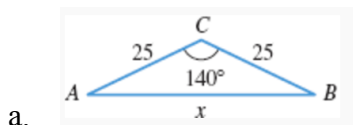


Heron's Formula for the area of a triangle is an application of the Law of Cosines.

Heron's Formula states that the area of a triangle ABC is given by $A = \sqrt{s(s-a)(s-b)(s-c)}$

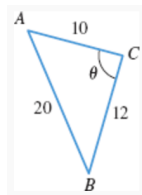
where $s = \frac{1}{2}(a+b+c)$ and is the **semiperimeter** of the triangle; that is, s is half the perimeter.

Example 1: Use the Law of Cosines to determine the indicated side x or angle θ .



$$x = \sqrt{25^2 + 25^2 - 2(25)(25)\cos 140^\circ}$$

$$= 46.98$$

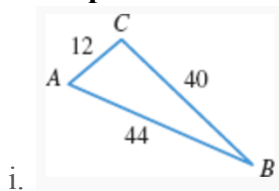


$$\cos \theta = \frac{10^2 + 12^2 - 20^2}{2 \cdot 10 \cdot 12}$$

$$\theta = \cos^{-1} \left[\frac{10^2 + 12^2 - 20^2}{2 \cdot 10 \cdot 12} \right]$$

$$\theta \approx 130.54^\circ$$

Example 2: Solve triangle ABC .



$$\angle A = \cos^{-1} \left[\frac{(44^2 + 12^2 - 40^2)}{(2 \cdot 44 \cdot 12)} \right]$$

$$= 62.96^\circ$$

$$\angle C = \cos^{-1} \left[\frac{(40^2 + 12^2 - 44^2)}{(2 \cdot 40 \cdot 12)} \right]$$

$$= 101.54^\circ$$

$$\angle B = \cos^{-1} \left[\frac{(40^2 + 44^2 - 12^2)}{(2 \cdot 40 \cdot 44)} \right]$$

$$= 15.5^\circ$$

iii. $a = 10, b = 12, c = 16$

$$\angle A = \cos^{-1} \left[\frac{(12^2 + 16^2 - 10^2)}{(2 \cdot 12 \cdot 16)} \right]$$

$$\approx 38.62^\circ$$

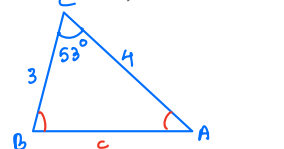
$$\angle C = \cos^{-1} \left[\frac{(10^2 + 12^2 - 16^2)}{(2 \cdot 10 \cdot 12)} \right]$$

$$= 92.87^\circ$$

$$\angle B = \cos^{-1} \left[\frac{(10^2 + 16^2 - 12^2)}{(2 \cdot 10 \cdot 16)} \right]$$

$$\approx 48.51^\circ$$

ii. $a_c = 3, b = 4, \angle C = 53^\circ$



$$c = \sqrt{3^2 + 4^2 - 2(3)(4)\cos 53^\circ}$$

$$c \approx 3.23 \text{ un}$$

$$\frac{c}{\sin 53^\circ} = \frac{3}{\sin A}$$

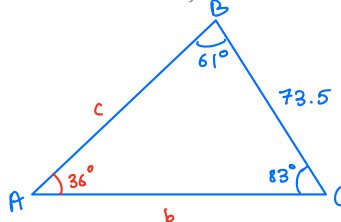
$$A = \sin^{-1} \left(\frac{3 \sin 53^\circ}{c} \right)$$

$$\angle A \approx 47.51^\circ$$

$$\angle B = 180^\circ - 53^\circ - \angle A$$

$$\angle B \approx 79.49^\circ$$

iv. $a = 73.5, \angle B = 61^\circ, \angle C = 83^\circ$



$$\frac{73.5}{\sin(36^\circ)} = \frac{b}{\sin(61^\circ)}$$

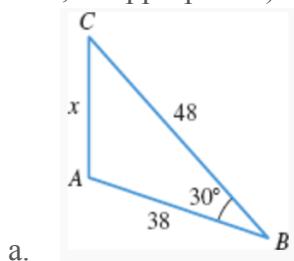
$$b = \frac{73.5 \sin(61^\circ)}{\sin(36^\circ)}$$

$$b \approx 109.37 \text{ un}$$

$$\frac{73.5}{\sin(36^\circ)} = \frac{c}{\sin(83^\circ)}$$

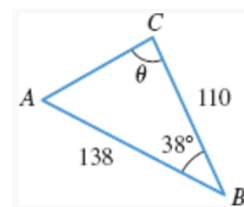
$$c = \frac{73.5 \sin(83^\circ)}{\sin(36^\circ)} \rightarrow c \approx 124.11 \text{ un}$$

Example 3: Find the indicated side x or angle. (Use either the Law of Sines or the Law of Cosines, as appropriate.)



$$x = \sqrt{38^2 + 48^2 - 2(38)(48)\cos 30^\circ}$$

$$= 24.26$$



$$AC = \sqrt{138^2 + 110^2 - 2(138)(110)\cos(38^\circ)}$$

$$\frac{\sin 38^\circ}{AC} = \frac{\sin \theta}{138}$$

$$\theta = \sin^{-1} \left[\frac{138 \sin 38^\circ}{AC} \right] \rightarrow \theta \approx 89.15^\circ$$

Example 4: Find the area of the triangle whose sides have the given lengths.

i. $a = 1, b = 2, c = 2$

$$s = \frac{1}{2}(1+2+2) = \frac{5}{2}$$

$$A = \sqrt{\frac{5}{2} \left(\frac{5}{2} - 1\right) \left(\frac{5}{2} - 2\right) \left(\frac{5}{2} - 2\right)}$$

$$= \boxed{0.97 \text{ unit}^2}$$

ii. $a = 11, b = 100, c = 101$

$$s = \frac{1}{2}(11+100+101) = 106$$

$$A = \sqrt{106(106-11)(106-100)(106-101)}$$

$$A \approx \boxed{549.64 \text{ unit}^2}$$

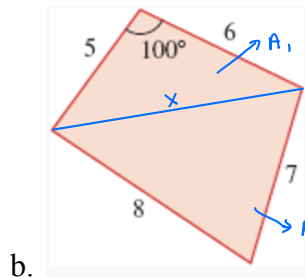
Example 5: Find the area of the shaded figure, rounded to two decimals.



$$s = \frac{1}{2}(3+4+6) = \frac{13}{2}$$

$$A = \sqrt{\frac{13}{2} \left(\frac{13}{2} - 3\right) \left(\frac{13}{2} - 4\right) \left(\frac{13}{2} - 6\right)}$$

$$A \approx \boxed{5.33 \text{ unit}^2}$$



$$x = \sqrt{5^2 + 6^2 - 2(5)(6)\cos(100^\circ)}$$

$$\approx 8.45 \rightarrow \text{sto } x$$

$$S_1 = \frac{1}{2}(x+5+6) = 9.73 \rightarrow \text{sto } y$$

$$S_2 = \frac{1}{2}(x+8+7) = \text{sto } z$$

$$A_{\text{TOT}} = A_1 + A_2$$

$$= \boxed{40.77 \text{ unit}^2}$$

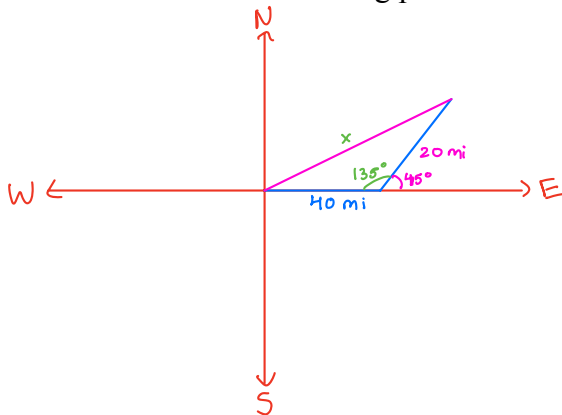
$$A_1 = \sqrt{y(y-5)(y-6)(y-x)}$$

$$= 14.77 \text{ unit}^2$$

$$A_2 = \sqrt{z(z-8)(z-7)(z-x)}$$

$$= 25.99 \text{ unit}^2$$

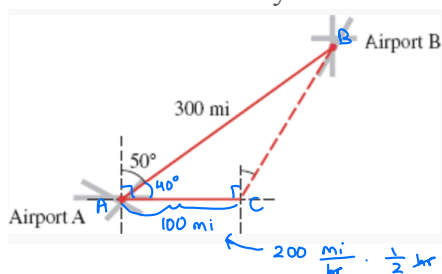
Example 6: A car travels along a straight road, heading east for one hour, then traveling for 30 min on another road that leads northeast. If the car has maintained a constant speed of 40 mi/h, how far is it from its starting position?



$$x = \sqrt{20^2 + 40^2 - 2(20)(40)\cos(135^\circ)}$$

$$\approx \boxed{56 \text{ mi}}$$

Example 7: Airport B is 300 mi from airport A at a bearing $N 50^\circ E$. A pilot wishing to fly from A to B has mistakenly flown due east at 200 mi/h for 30 min when the error is noticed.



(a) How far is the pilot from the desired destination at the time the error is noticed?

$$BC = \sqrt{300^2 + 100^2 - 2(300)(100)\cos(40^\circ)}$$

$$BC \approx 232.46 \text{ mi}$$

(b) At what bearing should the pilot head the plane to arrive at airport B?

$$\frac{\sin 40^\circ}{BC} = \frac{\sin C}{300}$$

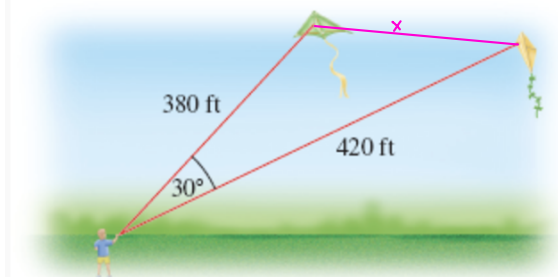
$$\angle C = \sin^{-1} \left[\frac{300 \sin 40^\circ}{BC} \right]$$

$$\text{Bearing} = 90^\circ - \angle C$$

$$\approx 33.95^\circ$$

$$N \approx 33.95^\circ E$$

Example 8: A child is flying two kites at the same time, with 380 ft of line let out to one kite and 420 ft let out to the other. It is estimated that the angle between the two lines is 30° . Approximate the distance between the kites.



$$x = \sqrt{380^2 + 420^2 - 2(380)(420)\cos 30^\circ}$$

$$= 210.63 \text{ ft}$$

Example 9: Land in downtown Columbia is valued at \$20 a square foot. What is the value of a triangular lot with sides of lengths 112, 148, and 190 ft?

$$S = \frac{1}{2} (112 + 148 + 190) = 225$$

$$A = \sqrt{225(225-112)(225-148)(225-190)}$$

$$A \text{ sq. ft.} \cdot 20 \frac{\text{dollars}}{\text{sq. ft.}}$$

$$\approx 16554.07 \text{ dollars}$$